

Heart Disease Prediction using Machine Learning

Task 1 Report

Student Information

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Task: Task 1 – Machine Learning Classification Project

1. Introduction

Heart disease is one of the leading causes of death worldwide. This project develops a machine learning model to predict whether a patient has heart disease using clinical attributes.

2. Objective

- Predict heart disease using machine learning.
- Compare Logistic Regression and Random Forest.
- Evaluate models using multiple metrics.
- Select the best-performing model.

3. Dataset

The project uses the UCI Heart Disease Dataset containing patient medical attributes. Target: 0 = No Heart Disease, 1 = Heart Disease.

4. Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

5. Methodology

Data Loading → Data Preprocessing → Exploratory Data Analysis → Train-Test Split (80:20) → Feature Scaling → Model Training → Cross Validation → Model Evaluation → ROC-AUC → Model Comparison.

6. Algorithms

Logistic Regression: Binary classification algorithm, simple, interpretable and effective.

Random Forest: Ensemble learning algorithm based on multiple decision trees.

7. Evaluation Metrics

Accuracy, Precision, Recall, F1 Score, Cross Validation, Confusion Matrix and ROC-AUC.

8. Results

Logistic Regression: Accuracy 85%, Precision 87%, Recall 84%, F1 Score 86%, ROC-AUC 0.93.

Random Forest: Accuracy 84%, Precision 84%, Recall 84%, F1 Score 84%.

9. Observations

Logistic Regression slightly outperformed Random Forest on this dataset. Cross-validation indicated stable performance. ROC-AUC of 0.93 indicates excellent discrimination.

10. Conclusion

The project successfully compared two machine learning algorithms for heart disease prediction. Logistic Regression achieved the best overall performance and was selected as the preferred model.

References

1. UCI Machine Learning Repository
2. Scikit-learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation